

JOHN WRIGHT

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PROFESSIONAL SUMMARY

M.S. Computer Science graduate with hands-on project experience in C/C++, Python, TypeScript, Unreal Engine 5, automated testing, CI, systems programming, and full-stack development. Built a dedicated-server multiplayer VR prototype with replicated interactions and server-authoritative gameplay logic, a deployed full-stack application with automated regression testing and GitHub Actions CI, and a concurrent C client-server application. Seeking software engineering, QA automation, Unreal Engine, VR/XR, simulation, and training-software development roles.

TECHNICAL PROJECT EXPERIENCE

Independent Software Development

Grand Junction, CO | May 2023 – Present

- **Multiplayer VR Prototype:** Built a dedicated-server multiplayer VR prototype in Unreal Engine 5 using Blueprints and the VR Expansion Plugin (VRE). Integrated VRE's physics-based gripping and held-object interactions, implemented replicated flashlight state, server-authoritative melee damage, enemy health and ragdoll death behavior, and PawnSensing-driven combat AI.
- **Top-Down GAS RPG:** Developed a top-down Unreal Engine 5 RPG project using C++, Blueprints, and the Gameplay Ability System. Implemented enemy combat, experience progression, save-slot persistence, ability loadouts, and upgradeable fireball, lightning, and arcane-shard abilities.
- **Full-Stack Task Manager:** Built and deployed a T3-stack TypeScript task-management application using Next.js, tRPC, Prisma, PostgreSQL/Neon, Tailwind CSS, and Auth.js email/password authentication with email verification. Implemented server-side task ownership and authorization through protected tRPC procedures, Prisma foreign keys, and user-scoped database queries.
- **Automated Testing & CI:** Created Playwright end-to-end and Python/Pytest API test suites covering login, task creation and validation, and rejection of unauthenticated API access. Configured GitHub Actions to run backend and browser regression tests against isolated Prisma/PostgreSQL schemas on pushes and pull requests.
- **Multi-Client Server:** Developed a concurrent client-server application in C using POSIX threads, UNIX domain sockets, mutex synchronization, signal handling, and thread lifecycle management.

Senior Capstone Project – Body Farm Database

Colorado Mesa University | Grand Junction, CO | January 2018 – May 2018

- Gathered requirements from researchers, translated operational needs into normalized relational database schemas, created tables, relationships, ER diagrams, data dictionaries, forms, reports, and queries to support data entry and reporting.

PROFESSIONAL EXPERIENCE

Audit Staff | Dalby Wendland | Grand Junction, CO

July 2019 – May 2020

- Prepared audit workpapers, analyzed financial data using advanced Excel, and communicated with clients to obtain and clarify supporting documentation.

EDUCATION

M.S., Computer Science — Stevens Institute of Technology, Hoboken, NJ

May 2025 | GPA: 3.934

B.S., Computer Information Systems; B.S., Public Accounting — Colorado Mesa University, Grand Junction, CO

December 2018 | GPA: 3.76

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, SQL

Unreal Engine & XR: Unreal Engine 5, Blueprints, C++, VR Expansion Plugin, Gameplay Ability System, Multiplayer Replication

Testing & Automation: Playwright, Selenium, Pytest, Python Requests, Postman, UI Testing, API Testing, Regression Testing, Page Object Model

Web & Backend: Next.js, React, tRPC, Prisma, Tailwind CSS, PostgreSQL, Authentication, HTTP APIs

Tools & Deployment: Git, GitHub, GitHub Actions, Diversion, Vercel, Neon